

Positioning, Impact, and Applications

Similar work in the literature, what is unique here, and where linear-map adapters and the shared hub save time, energy, and unlock local models.

1 · The landscape

The project sits between two research lines — representation-convergence science and embedding-space alignment engineering — both with active prior and contemporary work. The honest way to assess uniqueness is to map that landscape precisely.

work	what it showed	relation to this project
Kornblith et al. 2019 (CKA)	matched layers of separately trained vision networks are similar under CKA	Experiment A2 replicates the method on an LLM pair (GPT-2 vs Pythia) with a random-weights control
Lenc & Vedaldi 2015; Bansal, Nakkiran & Barak 2021 (stitching)	a single linear layer connects independently trained networks with small penalty	Experiment A3 replicates closed-form stitching layer by layer, adding the sample-starvation diagnosis
Git Re-Basin (Ainsworth et al. 2022)	after permutation alignment, independent runs lie in one loss basin	theoretical grounding for why weights differ while function converges
Platonic Representation Hypothesis (Huh et al. 2024)	representation geometries converge with model strength, across modalities	the framing hypothesis. This project confirms it in narrowed form AND answers one question its authors leave open (§2.3)
Gröger, Wen & Brbic 2026 (Aristotelian view)	after calibration, global convergence largely disappears while LOCAL neighbourhood agreement survives; similarity metrics are confounded by model width	independent confirmation of this project's central finding , published separately. Their calibration was applied here and the primary result passed (§2.3)
Convergence Without Understanding 2026	representations align while the computation over them diverges — convergence is a property of input encoding	structurally the same asymmetry measured here: spaces are readable out of but not writable into
vec2vec (2025)	translation between embedding spaces with no paired data, from geometry alone	stronger unsupervised cousin of the adapter; also the source of the privacy caveat
Text-to-Concept (Moayeri et al. 2023)	linear layers align vision models to CLIP space	closest academic precedent for 'linear map into a vision-language space'; research setting, not deployment
Drift-Adapter (2025)	lightweight adapters let a vector DB survive embedding-model upgrades	closest engineering precedent: same maths, applied to migration within one tier rather than across device tiers
EmbeddingAdapters; LlamaIndex linear query adapter	plug-and-play adapters between text-embedding spaces; transform one side only	same economic motive and the same one-sided asymmetry this project uses; not multimodal, not on-device

Table 1: the landscape. Two 2026 entries postdate the project's design and are treated as independent confirmation rather than influence.

2 · What is actually novel here

2.1 What is NOT novel

The primitive — a linear map between frozen embedding spaces — is not new, and two 2025 contemporaries (Drift-Adapter, EmbeddingAdapters) implement the same pattern. Claiming invention would be false. The accurate claim is different and still strong: this project independently arrived at, implemented and validated an instantiation the surveyed work has not covered, and did so with unusually complete experimental hygiene.

2.2 The combination that appears to be novel

	contribution	why the surveyed work does not cover it
a	Cross-tier, cross-vendor, multimodal pair	surveyed systems translate within one text tier, or align research models to CLIP offline. This bridges two production vision encoders from different vendors across a hardware boundary — an iPhone and a server sharing one live index
b	On-device deployment	the adapter ships as a 0.79 MB fp16 Core ML package on the Neural Engine. The surveyed precedents stop at Python
c	End-task gating with full baselines	validation is text-to-image Recall against both a ceiling and a chance baseline ($R@1 = 0.001$), with a pre-registered 90% gate that is reported as NOT met at 89.0%
d	Linearity proved by a failure, not assumed	a higher-capacity MLP trained on the true objective could not beat the matrix : 0.9183 against 0.9183, gain +0.0000, on two independent runs from a fresh 38,000-row corpus. Extra capacity buys nothing, so the map is not an approximation — it is the relationship. No surveyed engineering work includes this control
e	An N-encoder hub, not just pairs	the surveyed work maps between TWO spaces. This project builds one 512-d coordinate system that SEVEN encoders across four training objectives enter — 42 of 42 ordered pairs shared at 103.6% of bespoke per-pair hubs, with a component trained on one encoder serving encoders it never saw at 92.9–96.5%, including one of entirely independent lineage at 94.2%
f	Theory and application in one arc	the same project replicates the convergence science that justifies the engineering. Most work does one or the other

2.3 Where the project goes beyond its framing hypothesis

Describing this work as 'a small-scale confirmation of the Platonic Representation Hypothesis' now understates it in three specific ways.

what PRH leaves open	what this project measured
Huh et al. report cross-modal alignment at 0.16 of a possible 1.0 and ask explicitly whether the gap is noise or a real difference	part of it is neither: it is a COLLAPSED READOUT FRAME. Whitening a degenerate space — label-free and linear — lifts retrieval from 0.130 to 0.478, reaching 102.6% of an encoder trained contrastively on 1.2 billion pairs. Some 'missing' alignment in the literature may be a reading artifact rather than absent structure
directional asymmetry is noted as informative but not quantified	measured and explained: a collapsed space is readable OUT of at 95.1% and writable INTO at 24.8%, tracked by effective rank, with the obvious explanation tested and falsified
the hypothesis is stated as convergence toward a shared model of reality	narrowed to what survives measurement: agreement is LOCAL. Neighbourhood overlap runs hundreds of times chance and falls to roughly twice chance globally — confirmed on three metrics including the authors' own CKNNA, and independently by Gröger et al. (2026) on 204 model pairs

2.4 Why independent convergence with 2025–26 work is a feature

That unrelated teams — vector-DB migration, RAG tooling, and a separate ICML group — reached the same primitives and the same local-structure conclusion in the same period is evidence the pattern is real and timely. Arriving there independently, from theory, with a deployment target none of them addressed, is a legitimate contribution claim.

3 · Applications of the primitive

Any place where a cheap, local, or legacy encoder must interoperate with a canonical space is this pattern. The last two rows are hub-specific and were not available before Series G.

use case	how it is used	what it saves
Index migration / model upgrades	new encoder's vectors mapped into the legacy space (or queries mapped, documents untouched) while the corpus migrates lazily	re-embedding a 10M-item corpus (GPU-days) and index downtime; the upgrade becomes a 2 MB deploy
Cheap-encoder economics for RAG	a local MiniLM-class model embeds on device; the adapter lifts vectors into the premium API space	per-call embedding fees and 100–300 ms network latency per query
Edge-fleet ingestion	cameras, IoT and worker nodes run tiny encoders; adapters normalise all streams into one index	central GPU capacity and raw-media bandwidth (3 KB vector vs MB-scale image)
Encoder A/B testing	candidate encoder plus adapter searched against the production index before committing	building a parallel index per candidate
Cross-model RAG cache reuse	cached embeddings from a deprecated model stay queryable through an adapter	cold-start recomputation of historical corpora
Privacy-preserving indexing	sensitive media embedded on device; only adapted vectors leave	uploading raw personal media — with the vec2vec caveat that embeddings leak information and are not encryption
Federated multi-team search (hub)	teams with different embedding stacks each fit ONE map into a shared space rather than $N(N-1)$ pairwise bridges	measured, not projected: 42 of 42 pairs shared at 103.6% of bespoke; adding a team is one closed-form solve leaving every existing map untouched
Model federation instead of one generalist (hub)	K small domain experts, each strong in its own area, joined by a shared hub and served by components trained once	retraining a monolith to add a capability. A component trained on one encoder reaches 92.9–96.5% on encoders it never saw — the benchmark against a single large model is specified in Sources_and_Benchmarks.pdf and not yet run

Table 2: application catalog. Rows 7–8 rest on Series G measurements; the rest on Experiment B.

4 · Does it save time, energy, and improve local use?

Yes on all three — with the honest note that the figures below are order-of-magnitude engineering estimates stated with their assumptions, not measured on this project's hardware.

4.1 Compute and time

The adapter is computationally free at inference: a 512×768 matrix multiply is about 0.8 MFLOPs — roughly one five-thousandth of the MobileCLIP-S1 encode that precedes it, and vanishing next to SigLIP 2's tens of GFLOPs. The savings come from what it makes unnecessary: every image encoded once instead of twice, and no re-encoding on migration — for a 10-million-image corpus at ~50 ms per image server-side, that is GPU-days avoided.

4.2 Energy

Three effects. **The expensive encode never runs for indexing** — an on-device encode costs millijoules, a datacenter GPU encode amortised with cooling overhead sits at one to a few joules, one to two orders of magnitude more. **Data movement is avoided** — shipping a 3 KB vector instead of a 2–4 MB photo cuts per-item radio energy by about 99.9%, on both the phone and the network path. **One index instead of two** — a single shared graph halves the standing memory and search footprint. At personal scale the absolute saving is watt-hours; the point is architectural, and the same pattern at fleet scale puts those percentages onto real bills.

4.3 Improved local model use on mobile

This is the deepest effect: the adapter changes what a small on-device model is. Without it, a phone-side encoder is a silo — its vectors are useful only against other vectors it produced. With a 0.79 MB matrix the small model becomes a first-class citizen of the large model's ecosystem: photos indexed on the beach are searchable by the server's best text encoder the moment they sync, and the phone can search server content and the reverse. The cost of smallness is explicit — 88.4% of ceiling at R@1, 95.3% at R@5, 97.2% at R@10 — with a known escape hatch in server-side re-ranking of the top ten. It also future-proofs the tier: swapping the phone encoder for next year's better small model means refitting one matrix in seconds, with no corpus re-index and no architecture change.

The hub generalises this from two tiers to many. Where the adapter makes one small model a peer of one large one, the hub makes N models peers of each other at N maps rather than $N(N-1)$ — and the measured cost of centralising is nothing at all (103.6% of bespoke per-pair hubs). In short: adapters convert 'local model as a compromise' into 'local model as a peer with a published exchange rate', and the hub turns that exchange rate into a common currency.

4.4 Limits of the savings claim

The adapter does not reduce the cost of queries that were already server-side text encodes. It does not compress the index — vectors are 768-d either way. Its quality depends on the pair-fitting distribution (COCO-like photos here; domain shift to documents would need refitting). The hub adds one more: **width forces a choice** — narrow and components transfer, wide and transfer collapses (measured: 0.489 at 512 dimensions falling to 0.312 at 768), a trade-off that is real, reproduced, and currently unexplained. And the whole family inherits vec2vec's security lesson: adapted embeddings still carry recoverable information and must be protected like content, not like hashes.

Sources. Kornblith et al., ICML 2019 · Lenc & Vedaldi 2015 · Bansal, Nakkiran & Barak, NeurIPS 2021 · Ainsworth et al. 2022 (Git Re-Basin) · Huh et al., ICML 2024 (PRH) · Gröger, Wen & Brbic, ICML 2026 (Aristotelian view) · Convergence Without Understanding, 2026 · Jha et al. 2025 (vec2vec) · Moayeri et al., ICML 2023 (Text-to-Concept) · Drift-Adapter, 2025 · EmbeddingAdapters · LlamaIndex linear query adapter.